



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 02.08.2026

In addition to part I (General Handout for all courses appended to the Timetable) this portion gives further specific details regarding the course.

Course No.: PHY G541

Course Title: Physics of Semiconductor Devices

Instructor-in-charge: MANJULADEVI V.

1. Course Description:

This course will discuss the physics behind the semiconductors and the devices. The structure of any solid material plays an important role in their operation; the discussion will start with basic crystal structures and will cover the quantum theory of solids. In the later part of the course the discussion will be focused on the voltage current behavior of the devices. In the last part of the course, some special devices like solar cells, light emitting diodes etc. will be discussed.

2. Scope and objective of the course:

The objective of the course is to develop an understanding of the basic phenomenon of the semiconductor devices. The main content of the course is having theoretical description of these devices which is an essential requirement in order to understand the operational details of any device.

3. Text Book: Semiconductor Physics and Devices- Basic Principles by D.A. Neamen, McGraw Hill, 4th Edition, 2021.

4. Reference Book: Physics of Semiconductor Devices, S.M. Sze, K.K. Ng, Wiley India, 3e, 2007.

Introduction to Solid State Physics – Charles Kittel, Wiley, 8th Edition.

5. Course Plan:

Module Number	Lecture session/Tutorial Session.	Reference	Learning Outcome
Introduction Crystallography (Number of lectures: 4) L: 1-4	• Crystal Structures • Miller Indices • Imperfections and Impurities in Solids • Growth of Semiconducting Materials	TB1: Ch. 1 & Lecture slides	The students shall be able to learn basics about crystal structure, imperfections and impurities particularly applicable to semiconductors. Students will also learn about basic mechanism of growth of semiconducting materials
Energy Bands and Electrical Conduction (Number of lectures: 4) L: 5-8	• Energy bands and band gap. • Energy Band theory of semiconductors • Electrical Conduction in solids	TB1: Ch.3& Lecture slides	The students shall be able to learn the concept of Energy bands and bandgap. Students will learn about Energy band theory and Electrical Conduction concepts particularly applicable to semiconductors.
Charge Carriers and Fermi Energy Level	• Fermi Dirac Distribution • Charge carriers in Semiconductors	TB1: Ch. 4& Lecture slides	Students will be able to understand the role of



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(Number of lectures: 5) L: 9-13	• Intrinsic and Extrinsic semiconductor • Position of Fermi Energy level.		charge carriers and role of donors and acceptors in semiconductors.
Carrier Transport Phenomena (Number of lectures: 6) L: 14-19	• Drift and diffusion Current Density • Conductivity and Mobility • V-I characteristics • Hall Effect	TB1: Ch. 5& Ch. 6 (6.2, 6.3) Lecture slides	Students will develop an understanding of the carrier transport phenomena and the role of carrier drift and diffusion and the resulting current density and Mobility and also about Hall effect.
The pn Junction (Number of lectures: 6) L: 20-25	• Basic structure of pn junction • pn junction under different Bias conditions • Junction Capacitance	TB1: Ch. 7& Lecture slides	Students will develop an understanding about the basic structure of pn junction. Students will learn about the behavior of pn junction under different bias conditions and also understand the concept of junction Capacitance.
P-n Junction diode and Heterojunctions (Number of lectures: 7) L: 26-32	• Charge Flow in pn junction • pn junction current • diode current equation • Charge storage and Diode transients • Heterojunctions	TB1: Ch. 8& Lecture slides	Student will be able to know about the charge flow in pn junction and temperature effects, diffusion resistance etc. Students will learn about the IV characteristics of pn junction diodes. Students will learn about barriers resulting due to metal semiconductor ohmic contacts and about heterostructures.
Semiconductor Devices (Number of lectures: 8) L: 33-40	• Optical devices • Solar Cells • Photodetectors • Light emitting diodes • Laser diodes	TB1: Ch. 14& Lecture slides	Students will learn about the working principle of many semiconductor devices like solar cells, Photodetectors, LEDs and Laser diodes.



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6. Evaluation Scheme:

Component	Duration	Weightage (%)	Date & Time	Nature of component (Close Book/ Open Book)
Mid Semester Presentation	##	25		Project presentation followed by viva
Comprehensive Examination	180 Min.	40		Closed book/Open book
Tutorials/quizzes/ Assignments		35	***	Closed book/Open book

*** 5 Tutorial tests throughout semester to ensure continuous evaluation (4 best will be considered for 20% weightage); 2 take home assignments throughout the semester: 8% weightage; small quizzes during the daily classes whenever a submodule gets completed: 7% weightage

Individual projects will be assigned to each student in the beginning of the semester, they have to present their project (10% weightage this includes their interactions during the semester regarding their individual projects followed by a final presentation) and submit their project reports (10% weightage) followed by viva (5% weightage)

7. Chamber Consultation Hour: Monday (5:00-6:00 PM)

8. Notices: To be announced on NALANDA/Google classroom or by email.

9. Make-up Policy: No make-up for Tutorial tests. For other components, depending on the merit of the reason for missing the evaluative component.

10. Note (if any): It is a four unit course. The discussion related to the mentioned topics won't be confined to the books only. More readings from research papers and review articles will be expected from the students.

MANJULADEVI V.
(Instructor-in-charge)
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